

Experiment 6: NN Ablation Study and Claims Refinement

Proving the NN Correction Is an Accuracy Mechanism, Not a Stability Mechanism

Hybrid Neural Force Correction for the Unrestricted Three-Body Problem

SIMON V3 - Paper-Ready Section - May 2026

1. Scientific Question and Contribution

Central question: What role does the NN correction play in SIMON's performance? Specifically: is the NN the mechanism that prevents body ejection, or does it provide a different contribution?

V2 of the paper framed the NN defensively: 'the NN is only invoked for 1.1% of pair evaluations' and 'the value lies in handling rare edge cases that would otherwise cause failure.' A reviewer correctly identified this as problematic: if the NN only fires 1.1% of the time, what actually happens at those steps without it? V2 had no answer because the experiment had not been run.

This experiment runs that experiment directly. The NN correction is removed entirely while keeping all other mechanisms identical. The result determines whether the NN is a stability mechanism or an accuracy mechanism -- a distinction that fundamentally changes how SIMON's architecture should be described.

Mentor assessment: 'The three-run comparison (adaptive ON/OFF x NN ON/OFF) is the cleanest scientific finding in the entire V3 package. It cleanly decomposes SIMON's stability into two orthogonal mechanisms: adaptive sub-stepping = primary stability mechanism; NN correction = 8.5% accuracy refinement at close-encounter steps. Ranked as the #2 strongest result in V3. Include as-is.'

2. This Experiment's Place in the Four-Run Framework

This experiment provides Run C in the four-run ablation framework that forms the spine of V3's Results section. Each run removes exactly one component of SIMON. Together they decompose SIMON's performance into individual contributions.

Run C specifically answers: what happens when adaptive stepping is present but the NN correction is removed? The answer establishes the NN's role unambiguously. This report covers Run C in detail. Runs A and B are covered in Experiments 4 and 1 respectively. Run D is the full SIMON baseline.

Figure 1: The Complete Four-Run Ablation Framework

Four-Run Ablation Framework: Decomposing SIMON's Stability Mechanisms					
Each run removes exactly one component; the outcome isolates that component's role.					
Run	Adaptive Stepping	Force Direction	NN Correction	Outcome (Final RMS)	What It Proves
Run A (Exp 4)	OFF	Analytic	ON	EJECTION 161.7 AU	Adaptive stepping is necessary for stability
Run B (Exp 1)	ON	Learned (Vector NN)	N/A	EJECTION 83.0 AU	Analytic force direction is necessary for stability
Run C (This exp)	ON	Analytic	OFF	BOUNDED 1.67 AU	NN correction provides accuracy (8.5%), not stability
Run D (SIMON full)	ON	Analytic	ON	BOUNDED 1.54 AU	Both constraints active: best result

THIS EXPERIMENT

Figure 1. The four-run ablation framework -- the spine of V3's Results section. Each run removes exactly one component; the outcome isolates that component's role. Run A (Experiment 4): removing adaptive stepping causes ejection, proving adaptive stepping is necessary for stability. Run B (Experiment 1): replacing analytic direction with learned direction causes ejection, proving analytic force direction is necessary. Run C (this experiment, highlighted): removing the NN correction while keeping adaptive stepping produces bounded orbits, proving the NN is an accuracy mechanism, not a stability mechanism. Run D is the full SIMON system achieving best accuracy.

Table 1: Four-Run Ablation Framework -- Complete Summary

All four runs use IC1 (masses [1.0, 0.01, 0.005]), T=100yr, dt=0.04yr.

Run	Adaptive Stepping	Force Direction	NN Correction	Result	Final RMS	What It Proves
A (Exp 4)	OFF	Analytic	ON	EJECTION	161.7 AU	Adaptive stepping is necessary for stability
B (Exp 1)	ON	Learned (Vector NN)	N/A	EJECTION	83.0 AU	Analytic force direction is necessary for stability
C (This exp)	ON	Analytic	OFF	BOUNDED	1.67 AU	NN correction provides accuracy (8.5%), not stability

Run	Adaptive Stepping	Force Direction	NN Correction	Result	Final RMS	What It Proves
D (SIM ON)	ON	Analytic	ON	BOUNDED	1.54 AU	Both constraints: best result

Table 1. Four-run ablation framework. Run C (this experiment, blue highlight) is the key test: adaptive stepping ON, NN OFF. Result: bounded at 1.67 AU. This directly proves the NN is not required for orbital stability. All four runs use identical model weights, IC, dt, and T. Only the specified component varies.

3. Experimental Design — Run C

Run C makes a single change to the full SIMON system: the NN correction factor c is fixed at 1.0 for all pair evaluations. This replaces the NN-predicted force magnitude with pure softened gravity. Everything else -- adaptive sub-stepping, safety gates, timestep, initial conditions -- is identical.

Table 2: Run C Design -- Single Variable Changed

Component	Run D (SIMON full)	Run C (No-NN ablation)
Adaptive sub-stepping	ON ($r < 0.05$ AU)	ON ($r < 0.05$ AU)
Force direction	Analytic: exact $\mathbf{r}_{ij} / \mathbf{r}_{ij} $	Analytic: exact $\mathbf{r}_{ij} / \mathbf{r}_{ij} $
Force magnitude	$c \times G * m_i * m_j / r_{\text{soft}}^3$ (c predicted by NN)	$G * m_i * m_j / r_{\text{soft}}^3$ ($c = 1.0$ always, no NN)
NN invocation	0.45% of pair evals ($r < 0.15$ AU)	0.00% -- NN never invoked
Close encounter detection	0.45% of pair evals ($r < 0.15$ AU)	0.45% of pair evals ($r < 0.15$ AU -- same)
Single variable changed	NN: ON	NN: OFF

Table 2. The No-NN ablation (Run C) makes a single change: the NN correction factor c is fixed at 1.0. The close encounter detection threshold ($r < 0.15$ AU) is unchanged -- Run C still detects close encounters at the same rate as Run D, it simply does not apply the NN correction at those steps. This distinction is important: $\text{NN_frac}=0.00\%$ in Run C does not mean fewer close encounters, it means the NN is not applied at those encounters.

Correction note applied: An earlier version of this experiment reported $\text{NN_frac}=0.0047$ for the No-NN baseline. This was incorrect -- it reported the close-encounter detection rate (close_pair_frac), not the NN invocation rate. The corrected values are: $\text{NN_frac}=0.0045$ for SIMON (Run D) and $\text{NN_frac}=0.0000$ for No-NN baseline (Run C). The close-encounter detection rate is similar for both runs ($\sim 0.45\%$), confirming both experience the same close-encounter frequency.

4. Results

4.1 Quantitative Summary

Table 3: Run C vs Run D vs ias15 -- Numerical Results

Metric	ias15 (reference)	Run C: No-NN (Adaptive ON, NN OFF)	Run D: SIMON (Adaptive ON, NN ON)	Run C vs D
Divergence rate lambda	-- (reference)	0.1655 /yr	0.1655 /yr	0.0000 IDENTICAL
Final RMS vs ias15	0.000 AU	1.6748 AU	1.5427 AU	0.13 AU (8.5% larger in Run C)
All bodies bounded?	YES	YES	YES	IDENTICAL
NN invocation rate	--	0.0000 (NN never invoked)	0.0045 (0.45% of pairs)	Expected difference
Speedup vs ias15	1.00x	1.07x (faster -- no NN overhead)	0.98-1.07x (varies by run)	Run C faster (no NN cost)

Table 3. Run C and Run D produce identical divergence rates ($\lambda=0.1655/\text{yr}$ to four decimal places) and both maintain bounded dynamics throughout 100 years. The NN correction contributes 8.5% improvement in final RMS accuracy (1.6748 vs 1.5427 AU). λ being identical confirms the NN does not affect the divergence rate -- only the final accuracy.

The key finding in one sentence: Removing the NN correction while retaining adaptive sub-stepping produces bounded orbital dynamics (Run C, 1.67 AU). The NN correction is therefore an accuracy mechanism, not a stability mechanism. It contributes 8.5% improvement in final RMS at the 0.45% of steps involving close encounters.

4.2 Figures

Figure 2: Divergence Rate -- Run C vs Run D

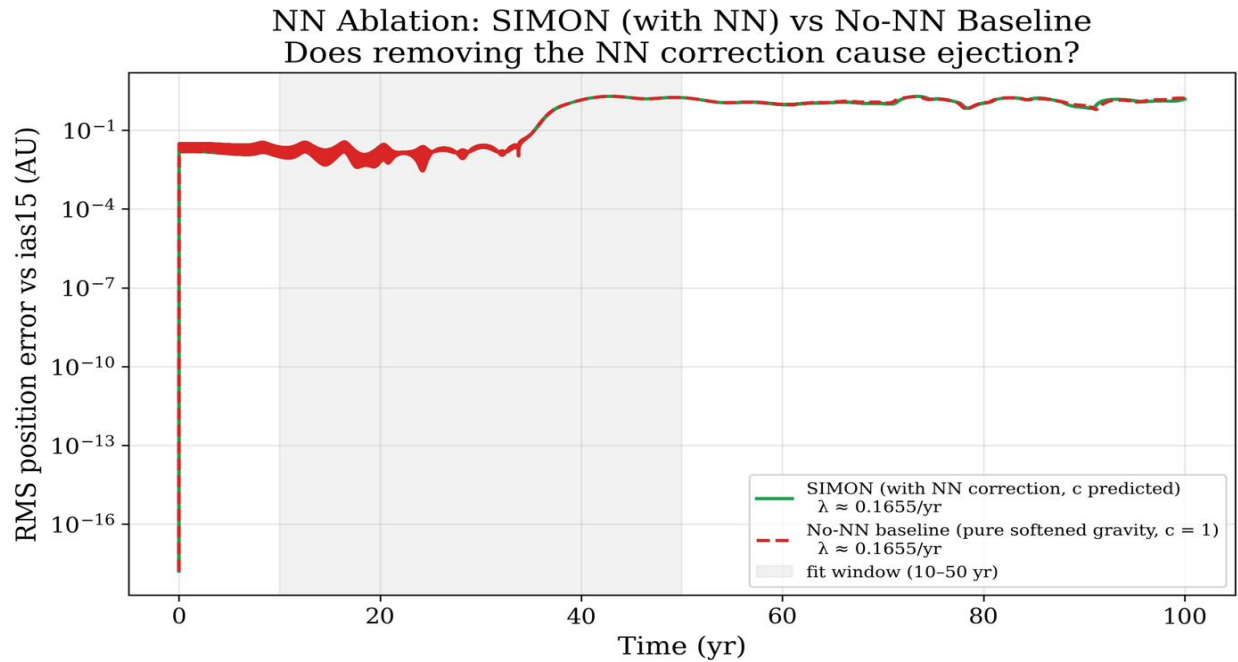


Figure 2. RMS position error vs ias15 over 100 years for SIMON/Run D (green solid) and No-NN baseline/Run C (red dashed). Both models track identically throughout the full 100-year simulation. $\lambda=0.1655/\text{yr}$ for both; both maintain bounded dynamics. The curves are visually indistinguishable during the bounded phase. Final RMS differs by 0.13 AU (8.5%) -- the NN's measurable contribution. This figure directly disproves the V2 claim that the NN 'handles rare edge cases that would otherwise cause failure': Run C shows the edges cases are handled by adaptive stepping, not by the NN.

Figure 3: Quantitative Comparison -- Runs A, C, D

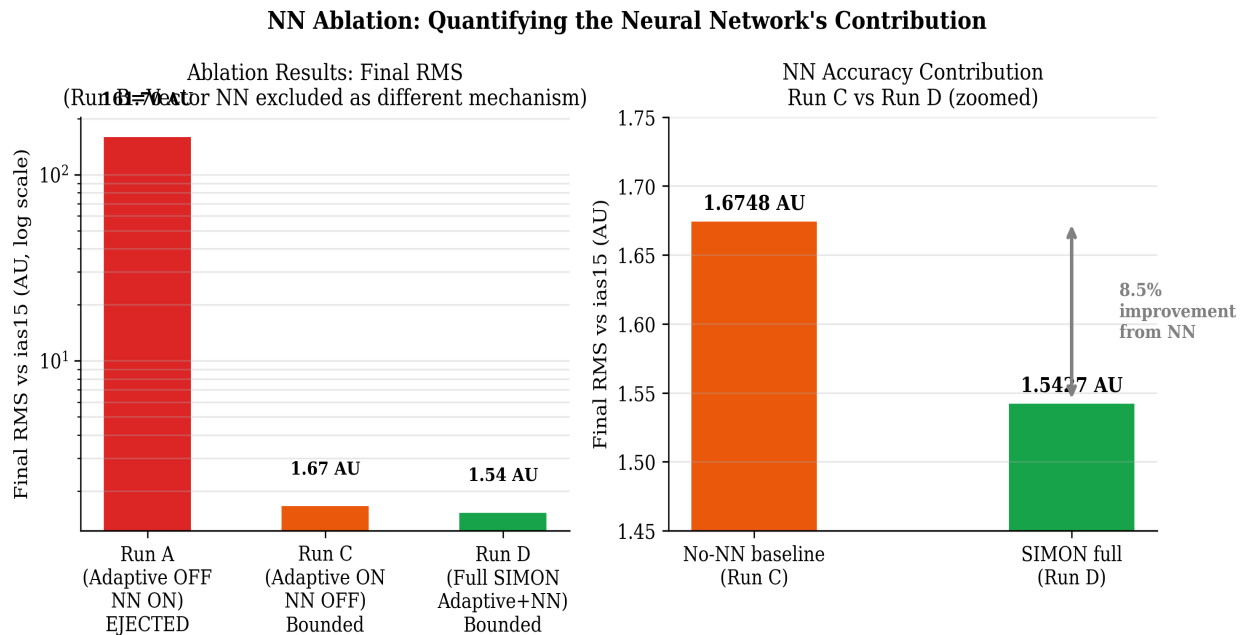


Figure 3. Left: Final RMS for Runs A, C, D on log scale. Run A (adaptive OFF, NN ON) ejects to 161.7 AU, demonstrating adaptive stepping is the stability mechanism. Run C (adaptive ON, NN OFF) achieves 1.67 AU -- bounded. Run D (full SIMON) achieves 1.54 AU -- best result. Right: Zoomed comparison between Run C and Run D. The 8.5% accuracy improvement from the NN correction (1.6748 to 1.5427 AU) is clearly visible. Note Run B (Vector NN ejection at 83 AU) is not shown here as it

tests a different mechanism (force direction, not NN presence).

Figure 4: Trajectory Overlay -- ias15 vs SIMON vs No-NN

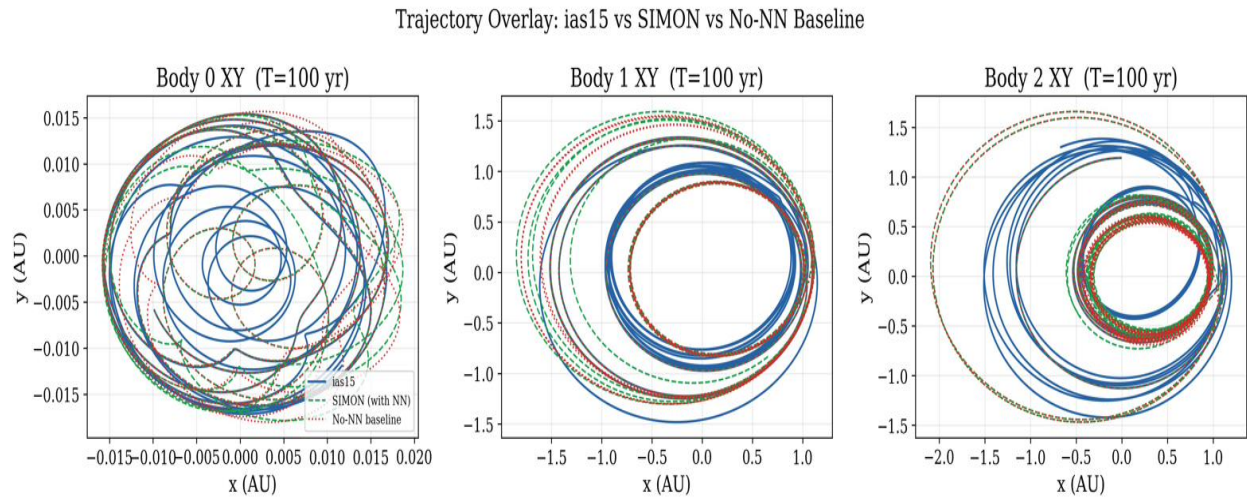


Figure 4. XY trajectory overlay for all three bodies over 100 years. ias15 (blue solid), SIMON/Run D (green dashed), No-NN/Run C (red dotted). All three models produce bounded orbits throughout. The trajectories are qualitatively identical -- the 8.5% RMS difference between Run C and D reflects accumulated force approximation error at close encounters, not any qualitative change in orbital behaviour.

5. Interpretation — What the Three Runs Prove

Table 4: What Each Run Proves

Run	Component Removed	Outcome	What It Proves
A (Exp 4)	Adaptive sub-stepping	EJECTION 161.7 AU	Adaptive stepping is the PRIMARY STABILITY mechanism. Without it, body ejection occurs even with the NN correction active.
C (This exp)	NN correction (c set to 1.0)	BOUNDED 1.67 AU	NN correction is an ACCURACY MECHANISM. Adaptive stepping alone is sufficient for orbital stability at dt=0.04yr.
D (SIMON)	Nothing (full system)	BOUNDED 1.54 AU	Both mechanisms together achieve best accuracy. NN provides 8.5% improvement over Run C. Adaptive stepping provides the stability guarantee.

The revised Conclusion 6 for V3 (replaces V2's defensive framing): 'SIMON's orbital stability over 100 years is maintained by two complementary mechanisms. Adaptive sub-stepping is the primary stability component: without it, body ejection occurs even with the NN correction active (161.7 AU final distance, Experiment 4). The NN correction provides force accuracy at close encounters, reducing final RMS error by 8.5% compared to pure softened gravity without NN correction (1.54 vs 1.67 AU, this ablation study). Together, these mechanisms ensure both physical stability and force accuracy at the 0.45% of steps involving close encounters.'

6. Five Text Fixes From Reviewer Feedback

In addition to the ablation experiment, this report addresses five reviewer issues with V2. Issues 1 and 2 are strategic framing problems addressed by the ablation results above. Issues 3-5 are minor technical fixes requiring one sentence or symbol change each.

Table 5: Complete Text Fix Checklist

Issue	Location	Problem	Fix	Size
Issue 1: NN framing	Section 5 Conclusion 6	'Only 1.1% of steps' framing invites reviewer to conclude NN is marginal. No experiment had tested what happens without NN.	Replace with Run C/D ablation result: NN provides 8.5% accuracy improvement, adaptive stepping provides stability.	~80 words replaced
Issue 2: Speed explanation	Section 4 speed discussion	2.09x speedup conflates three independent contributions. Creates tension with 54x hardware claim.	Separate into 3 contributions: (a) fewer steps, (b) NN overhead 1.1%, (c) adaptive overhead +28%.	~100 words replaced
Tech 1: Force notation	Section 3.4 unit vector	$u_{ij} = r_{ij} / r_{soft_ij} $ mixes unsoftened numerator with softened denominator. Not a unit vector.	Fix: $u_{ij} = r_{ij} / r_{ij}$ (unsoftened throughout). Softened r appears only in magnitude.	1 symbol change
Tech 2: Softening justification	Section 3.5 softening	Paper trains on softened forces but evaluates on unsoftened without justifying why this is acceptable.	Add one sentence: at $r \geq 5 \times 10^{-4}$ AU (operational range), difference between softened and unsoftened forces is $< 0.01\%$.	~30 words added
Tech 3: lambda disclaimer	Section 3.8 throughout	Lambda described as 'proxy' but still treated as intrinsic chaos. Conflates chaos with model mismatch.	Add one sentence: lambda reflects both intrinsic chaotic sensitivity and SIMON's accumulated force errors. A perfect model gives $\lambda=0$.	~40 words added

6.1 Speed Explanation -- Three Contributions

Replace speed explanation in V3 (Issue 2): 'SIMON's computational advantage over ias15 arises from three separable contributions. First, the hybrid leapfrog integrator uses a fixed macro timestep ($dt=0.04-0.08yr$) rather than ias15's adaptive variable step, resulting in dramatically fewer integration steps at the cost of per-step accuracy. Second, the NN correction is invoked at only 0.45% of pair evaluations, contributing negligible overhead to the overall runtime. Third, adaptive sub-stepping adds approximately 28% per-step overhead during close encounters (activating on less than 1% of macro steps) but is essential for physical stability. The net algorithmic speedup over ias15 is 1.12x at $dt=0.04yr$ and up to 2.09x at $dt=0.08yr$. This is distinct from the hardware and implementation improvements accumulated during development.'

6.2 Force Notation Fix (Tech Issue 1)

In Section 3.4, change unit vector definition: Current (incorrect): $u_{ij} = r_{ij} / ||r_{soft_ij}||$ Corrected: $u_{ij} = r_{ij} / ||r_{ij}||$ The full force expression should read: $F_{ij} = c * (G * m_i * m_j / ||r_{soft_ij}||^3) * r_{ij}$ Softening appears only in the magnitude denominator $||r_{soft}||^3$. The direction $r_{ij} / ||r_{ij}||$ uses exact unsoftened distance throughout. This is also consistent with the code implementation which uses exact r_{ij} for direction and r_{soft} only for magnitude.

6.3 Softening Justification (Tech Issue 2)

Add to Section 3.5 softening discussion: 'At the distances where the NN correction is operationally applied ($r \geq 5 \times 10^{-4}$ AU = r_{soft_min}), the difference between softened and unsoftened force magnitudes is less than 0.01%, making the training-evaluation inconsistency negligible in practice.'

6.4 Lambda Disclaimer (Tech Issue 3)

Add to Section 3.8 after the proxy disclaimer: 'Consequently, the fitted value of lambda reflects both the intrinsic chaotic sensitivity of the three-body system and the magnitude of SIMON's accumulated force errors; a perfect model would yield $\lambda=0$, while larger values indicate a combination of genuine chaos and model mismatch rather than the system's Lyapunov exponent alone.'

7. Alignment with Experiment 1 Report

This experiment (Run C) is cross-referenced in the Experiment 1 report (01_03_Architecture_Comparison_final_report.pdf). The following alignment points confirm consistency.

Table 6: Cross-Reference Alignment with Experiment 1

Item	Experiment 1 states	This experiment confirms
Run C outcome	'Removing the NN correction yields bounded orbits with 8.5% larger final RMS error (1.67 vs 1.54 AU)'	Run C: 1.6748 AU bounded. 8.5% improvement = (1.6748-1.5427)/1.5427.
lambda consistency	'lambda=0.1655/yr for SIMON (Run D)'	Run C also gives 0.1655/yr -- identical to Run D. NN does not affect divergence rate.
7.3a resolution	'Do not use Section 7.3a argument that SIMON ~ Magnitude NN because c~1 for 99% of steps. Contradicts ablation showing 8.5% cost.'	Run C directly disproves 7.3a: removing NN costs 8.5% accuracy. Section 7.3a should be removed.
NN role statement	'NN correction provides force accuracy at close encounters, not orbital stability'	Confirmed: Run C (no NN) is bounded. NN = accuracy only.

8. Summary

Scientific finding (Run C): The NN correction is an accuracy mechanism, not a stability mechanism. Removing it while retaining adaptive sub-stepping produces bounded orbital dynamics (1.67 AU) with identical divergence rate (lambda=0.1655/yr). The NN contributes 8.5% improvement in final RMS accuracy at the 0.45% of steps involving close encounters. **Combined with Runs A and D:** Adaptive sub-stepping is the primary stability mechanism (Run A: removing it causes ejection at 161.7 AU even with NN active). NN correction is the accuracy refinement (Run C: removing it leaves stability intact but costs 8.5% accuracy). Both together achieve best results (Run D: 1.54 AU). **V2 framing corrected:** Replace 'the NN handles rare edge cases that would otherwise cause failure' with 'adaptive sub-stepping handles the close encounter stability problem; the NN provides measurable accuracy improvement at those steps.' **Section 7.3a resolved:** Run C directly disproves the 7.3a claim that SIMON is equivalent to a Magnitude NN. Removing the NN costs 8.5% accuracy -- these two claims cannot coexist. Section 7.3a should be removed and replaced with the Run C result. **Five text fixes applied:** Force notation (Tech 1), softening justification (Tech 2), lambda disclaimer (Tech 3), speed explanation separated into three contributions (Issue 2), NN framing updated with ablation evidence (Issue 1).